

Visualization of Modal Irradiance Patterns in an Optical Fiber

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Abstract—A laboratory exercise that allows the student to combine theory, numerical computation methods, and experimental observation to learn more about electromagnetic modes and modal patterns is presented.

I. INTRODUCTION

THE electromagnetic analysis of a waveguiding structure is fundamentally the solution of the structure's modes and their propagation properties. However, the mathematical solution of the modes is often not enough to convey a "feel" for what a mode is or looks like. This paper presents a stimulating way of combining electromagnetic theory, numerical methods, computer visualization, and experimental observation.

Since the modes of an optical fiber can be seen by the human eye, it is an ideal means of teaching about modes. The students can visually compare their calculated mode patterns with those actually observed in a fiber. The computer visualization of these modal patterns has not usually been done on the undergraduate level since it requires: 1) the solution of two nonlinear simultaneous equations (one which contains Bessel functions of two kinds) and 2) sophisticated computer graphics.

The application of Maxwell's equations to the optical fiber structure results in characteristic equations which determine the properties of the fiber's propagating modes. These equations can be simplified if the fiber's core and cladding refractive indexes are nearly equal [1]. The modes obtained from the simplified equations are referred to as linearly polarized (LP) modes [2]. ([2] is a senior undergraduate/first-year graduate level textbook which contains an excellent discussion of optical fiber modes.)

To determine the propagation parameters (u and w) associated with the LP modes, two nonlinear simultaneous equations must be solved. The first is the characteristic equation for the LP_{ml} modes

$$(1/u)[J_m(u)/J_{m-1}(u)] = -(1/w)[K_m(w)/K_{m-1}(w)] \quad (1)$$

where J_m is the Bessel function of the first kind of order m , and K_m is the modified Bessel function of the second kind of

order m . The second is the normalized wavevector equation

$$u^2 + w^2 = v^2. \quad (2)$$

[1] and [2] contain the derivation and a complete discussion of these equations.

The computer software package *Mathematica* [3] provides all of the tools necessary to solve these two equations simultaneously and plot them graphically. The two equations are solved numerically using the Newton-Raphson method. Having determined the necessary parameters, the modal irradiance pattern is calculated and then graphically displayed. *Mathematica* contains a display function called *DensityPlot* [3, pp. 136–137], which is especially useful for this display. In a density plot, the brightness of a pixel is proportional to the value of the plotted function at that point. This allows the student to directly compare the observed and calculated patterns. Calculated irradiance plots of the LP_{21} mode, LP_{02} mode, and a combination of the LP_{21} and LP_{02} modes are displayed in Fig. 1. The commented *Mathematica* script used to generate them is given in the Appendix.

To observe the irradiance patterns in the lab, a fiber designed to be single mode at a freespace wavelength $\lambda_0 = 1300$ nm is excited with a helium-neon laser ($\lambda_0 = 633$ nm). Four LP modes (LP_{01} , LP_{11} , LP_{02} , LP_{21}) can propagate in this fiber at $\lambda_0 = 633$ nm. To see the modal structure of the fiber, the exit end of the fiber is imaged on a wall or screen using a 40X microscope objective lens [4]. The student can then directly compare the calculated and observed patterns.

Each mode can be uniquely launched by varying the position and incident angle of the light on the fiber core. Exciting each mode uniquely is often difficult to accomplish in one lab period; it is more likely that some combinations of two modes will be observed. For the patterns consisting of mode combinations, the fraction of the total irradiance associated with each mode can be quantified by summing the previously calculated modal patterns with individual "weights," and adjusting the "weights" until the calculated pattern matches the observed pattern. Quantifying these mode combinations is an important exercise, since an understanding of the distribution of power among the modes of a multimode fiber is important for many applications.

The redistribution of power among the modes of the fiber when it is bent can be easily demonstrated by simply bending

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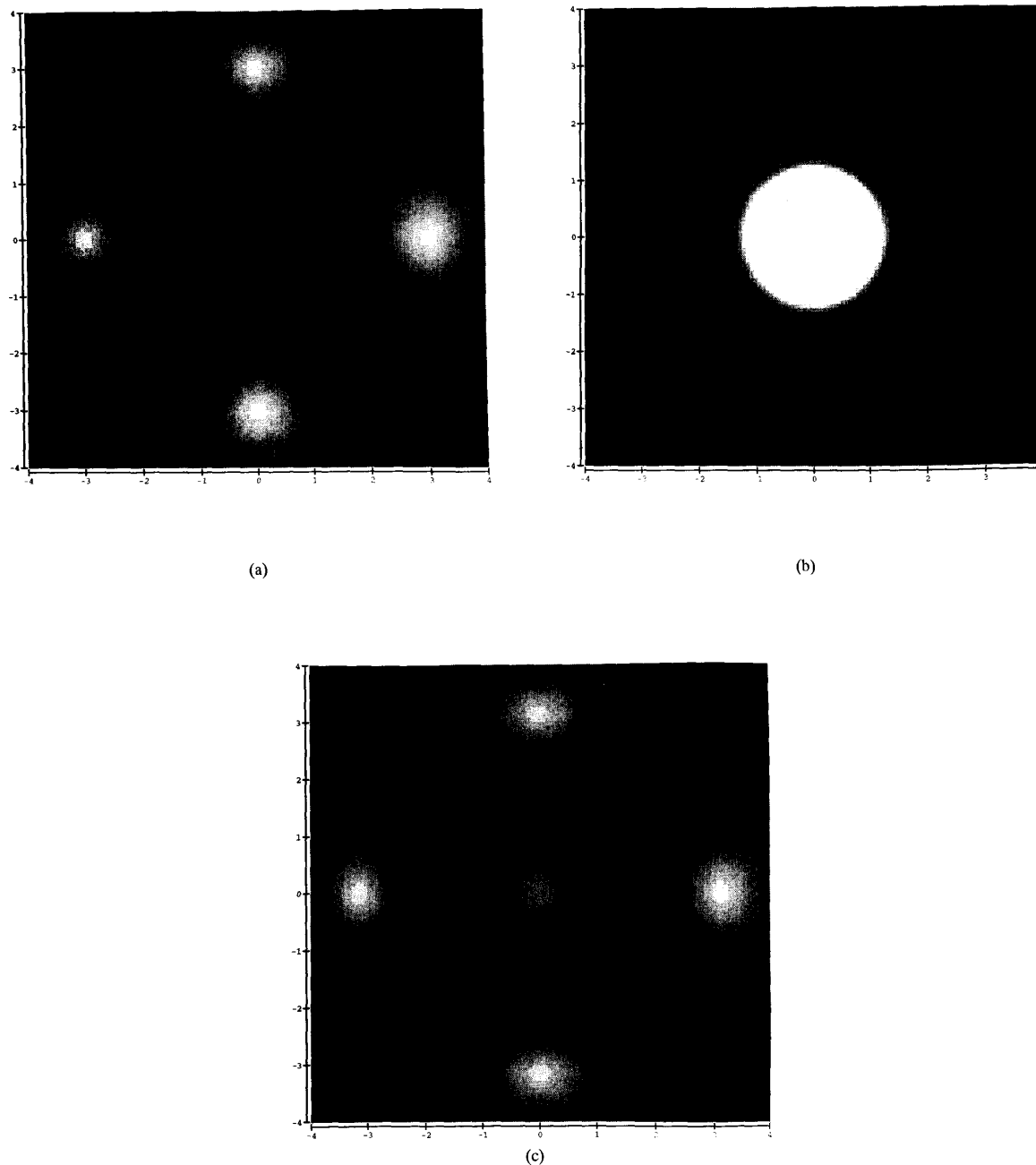


Fig. 1. Calculated density plots of the normalized irradiance of the following modes. (a) LP_{21} . (b) LP_{02} . (c) A combination of 80% LP_{21} and 20% LP_{02} .

the fiber and watching the image; it can be quantified by looking at the change of the "weights" of each mode in the composite pattern. This is a relevant exercise since one means of approximating a steady-state power distribution among the modes, necessary for meaningful attenuation measurements, is to bend the fiber [5].

In conclusion, this combined computer and laboratory exercise provides students with the opportunity to: 1) derive the field equations and the characteristic equations from first

principles; 2) solve for the parameters of the modes and plot them; and 3) visually compare the modes observed in the laboratory with those calculated. This "tying together" has been quite rewarding for my students in the past.

APPENDIX-COMMENTED MATHEMATICA SCRIPT

Input statements are indicated by boldface type. Comments are enclosed in (* and *).

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FindRoot [{u^2 + w^2 == 19.2721 (*v^2 = 19.2721*),
  u BesselJ[1,u]/BesselJ[2,u] == -w BesselK[1,w]/BesselK[2,w]
  (* Characteristic equation for LP_21 mode *)},
{u,4.0 (* begins search around u = 4.0 *)},
{w,1.0 (* begins search around w = 1.0 *)}]
{u → 4.03311, w → 1.73381}
(* FindRoot employs the Newton-Raphson method to solve the two equations within the first set of braces *)
u1 = 4.03311 (* assigns the value for u to a constant *)
4.03311
kappa = u1/4.0 (* the transverse propagation constant, radius of the core is 4.0 micrometers *)
1.00828
f21[x_,y_] := (BesselJ[2,kappa Sqrt[x^2 + y^2]]
  Cos[2 ArcCos[x/Sqrt[x^2 + y^2]]])^2
(* Normalized Irradiance for LP_21 mode, proportional to [J_2(kappa r) Cos(2 theta)]^2 *)
DensityPlot[f21[x,y], {x,-4,4}, {y,-4,4}, PlotPoints → 90, Mesh → False]
(* Normalized Irradiance of LP_21 mode *)
kappa02 = 1.05514
(* transverse propagation constant for the LP_02 mode, calculated previously *)
f02[x_,y_] := (BesselJ[0,kappa02(Sqrt[x^2 + y^2])])^2
(* Normalized Irradiance for LP_02 mode, proportional to [J_0(kappa r)]^2 *)
DensityPlot[f02[x,y], {x,-4,4}, {y,-4,4}, PlotPoints → 90, Mesh → False]
(* Normalized Irradiance of LP_02 mode *)
DensityPlot[0.8 f21[x,y] + 0.2 f02[x,y], {x,-4,4}, {y,-4,4}, PlotPoints → 90, Mesh → False]
(* Normalized Irradiance of a combination of 80% LP_21 mode and 20% LP_02 mode *)

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REFERENCES

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